

SapBERT Classifier Comparison

Sections 3.4 and 3.5 | Complete research experiment comparison

Based on the latest uploaded triage_classifier.py. Its SapBERT encoder, data preparation, patient-feature preprocessing, grouped split and Logistic Regression / Hist Gradient Boosting settings are retained. Original 768-D tests and Random Forest are explicitly added comparisons; the uploaded file remains unchanged.

Dataset: 821 synthetic cardiac records with existing generated clinical concepts. The first recorded concept for each complaint was matched to cardiac_english_2252.csv. Split: 609 training and 212 test records; no exact concept string occurs in both sets. All comparisons use these same test records.

Encoder: cambridgeltl/SapBERT-from-PubMedBERT-fulltext. CLS pooling, L2 normalization and 64-token truncation produce 768-D embeddings. Training-only PCA reduces these to 64 dimensions and retains 90.7% of variance. Fusion adds 24 patient features (792 or 88 total features).

Text-only comparison

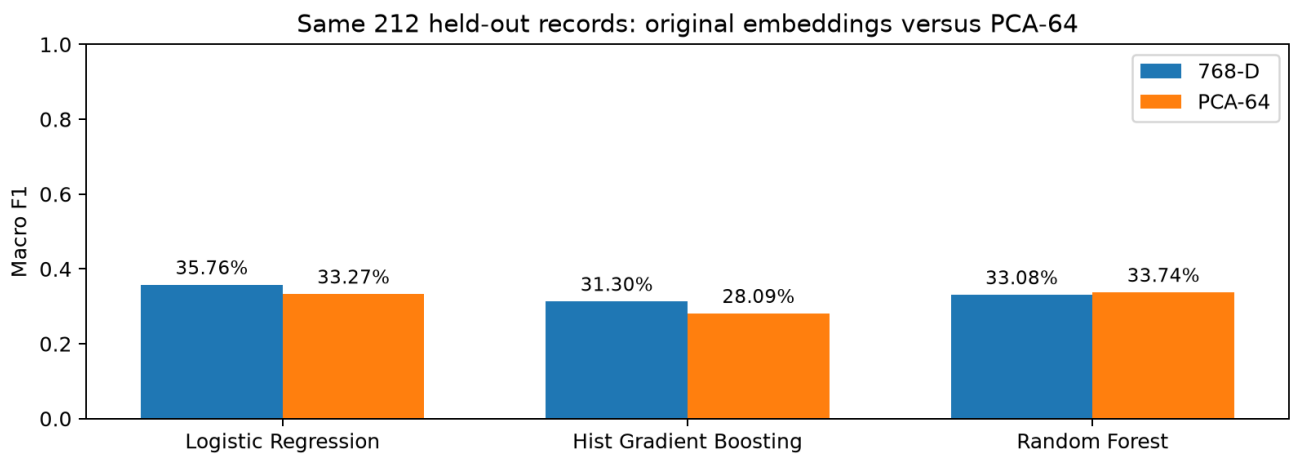
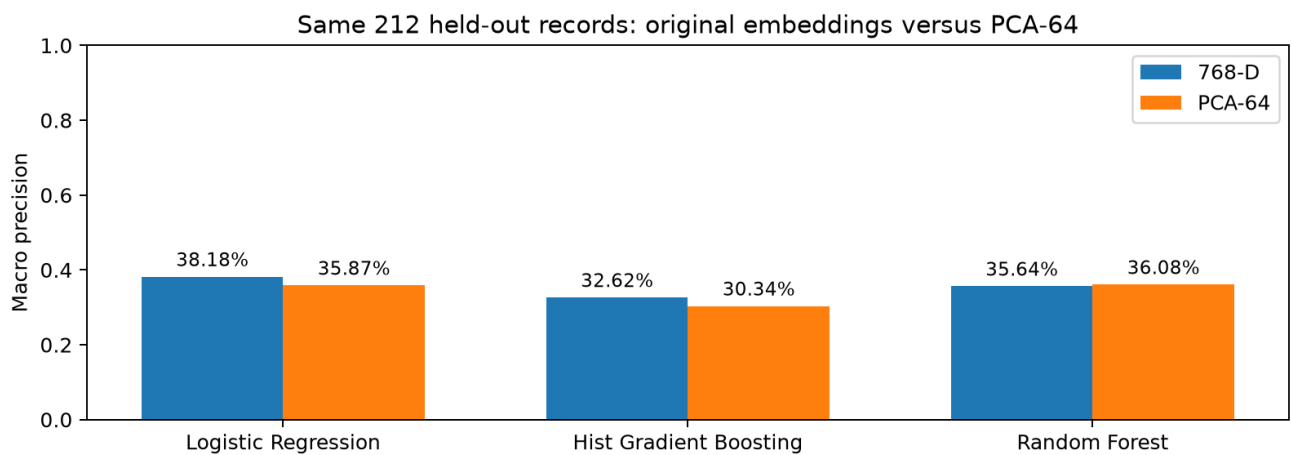
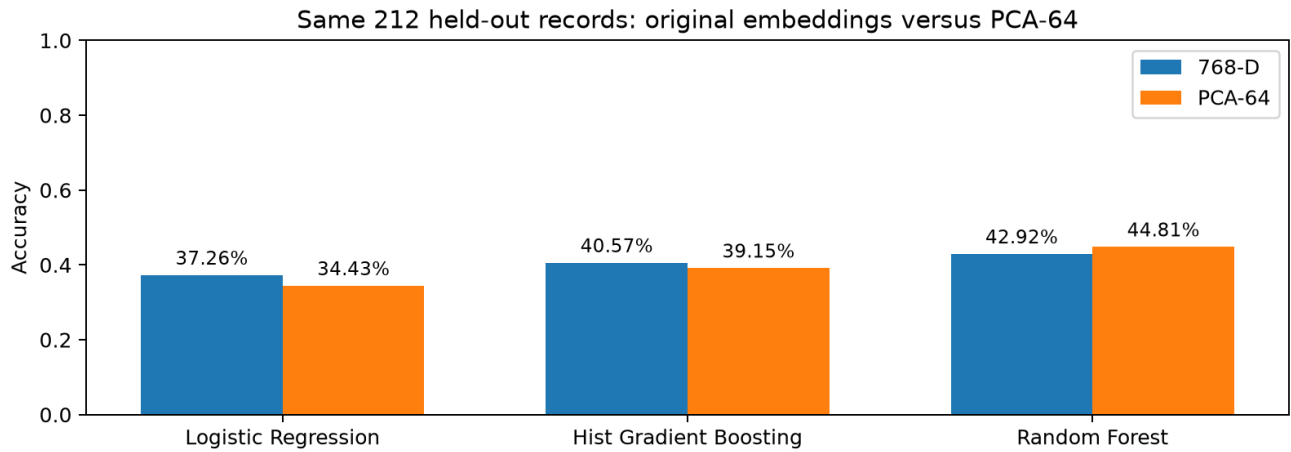
Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	37.26%	38.18%	35.76%
768-D	Hist Gradient Boosting	40.57%	32.62%	31.30%
768-D	Random Forest	42.92%	35.64%	33.08%
PCA-64	Logistic Regression	34.43%	35.87%	33.27%
PCA-64	Hist Gradient Boosting	39.15%	30.34%	28.09%
PCA-64	Random Forest	44.81%	36.08%	33.74%

Text plus patient features

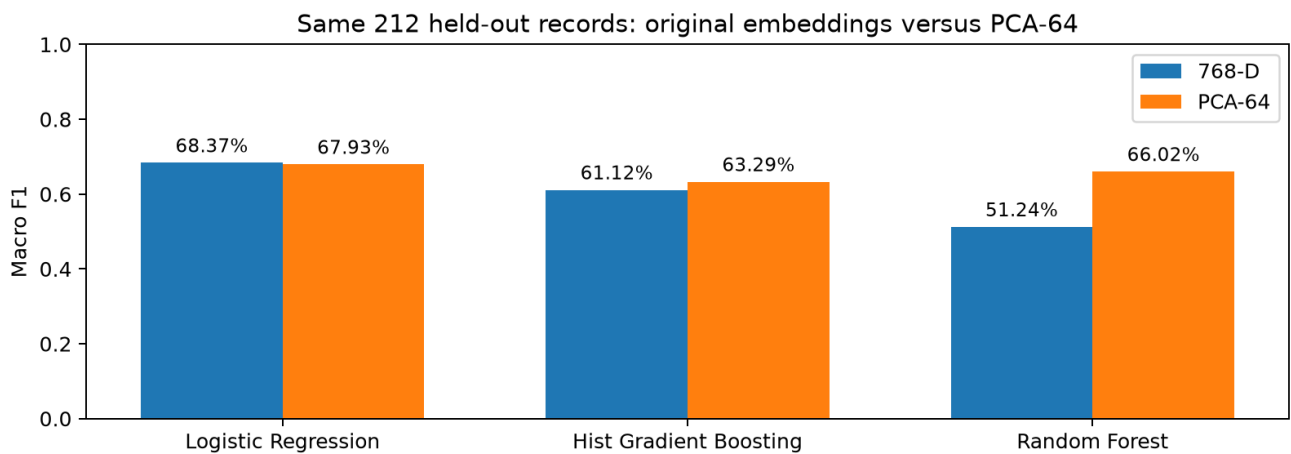
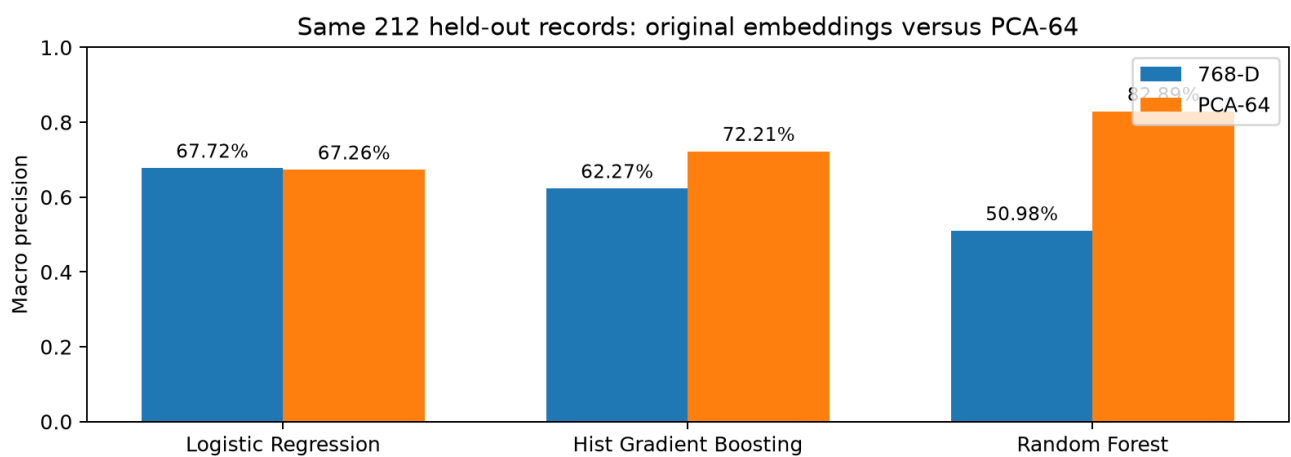
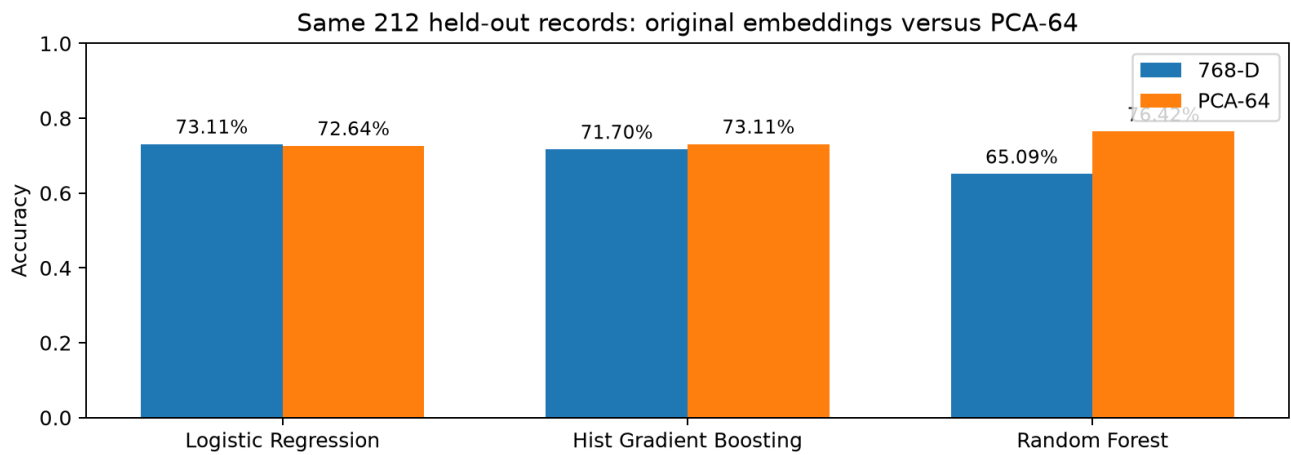
Text size	Classifier	Accuracy	Precision*	F1*
768-D	Logistic Regression	73.11%	67.72%	68.37%
768-D	Hist Gradient Boosting	71.70%	62.27%	61.12%
768-D	Random Forest	65.09%	50.98%	51.24%
PCA-64	Logistic Regression	72.64%	67.26%	67.93%
PCA-64	Hist Gradient Boosting	73.11%	72.21%	63.29%
PCA-64	Random Forest	76.42%	82.89%	66.02%

*Precision and F1 are macro-averaged. Scores describe this concept-based experiment; they are not the previous PDF's MPNet/English-translation experiment. Values are rounded for display; underlying CSV and JSON files retain full precision.

Text-only performance

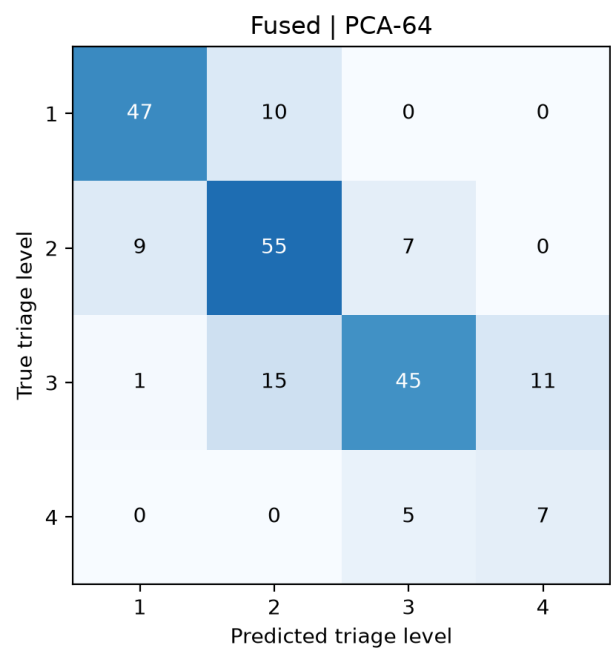
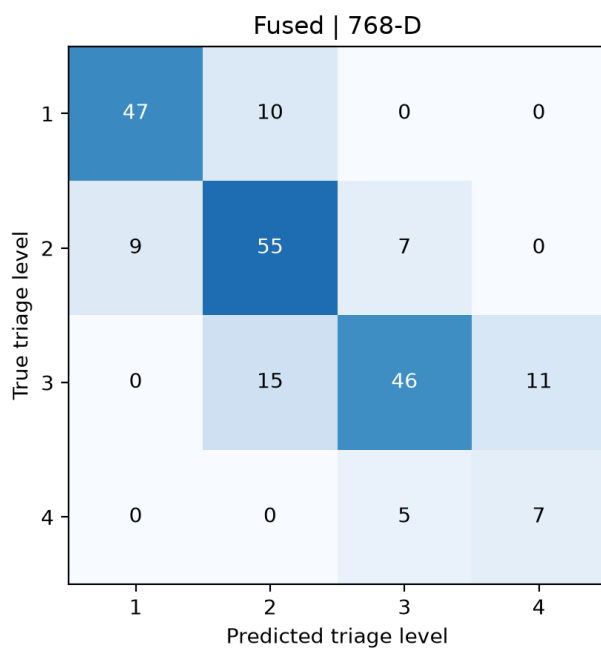
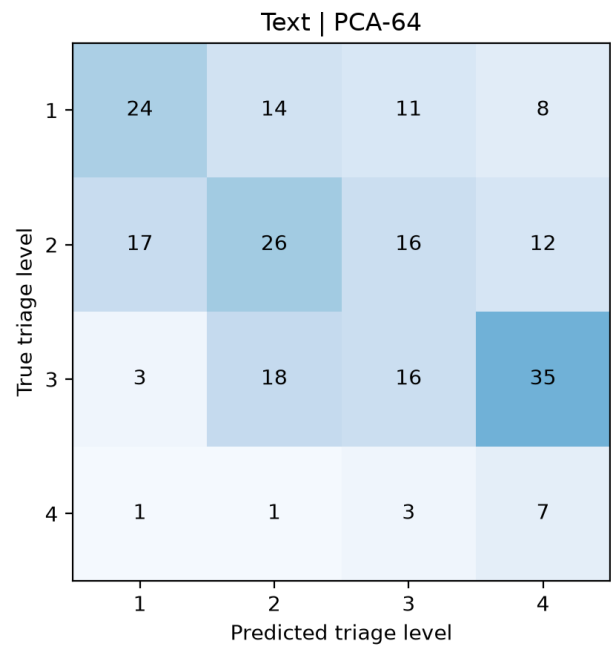
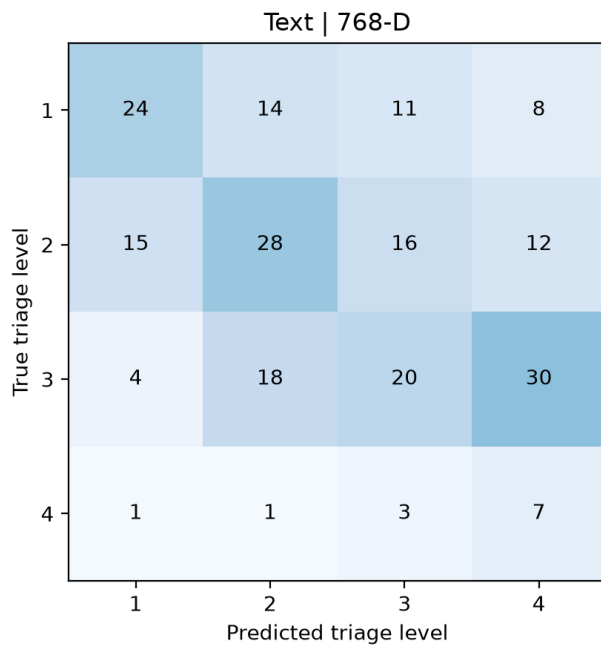


Performance with patient features



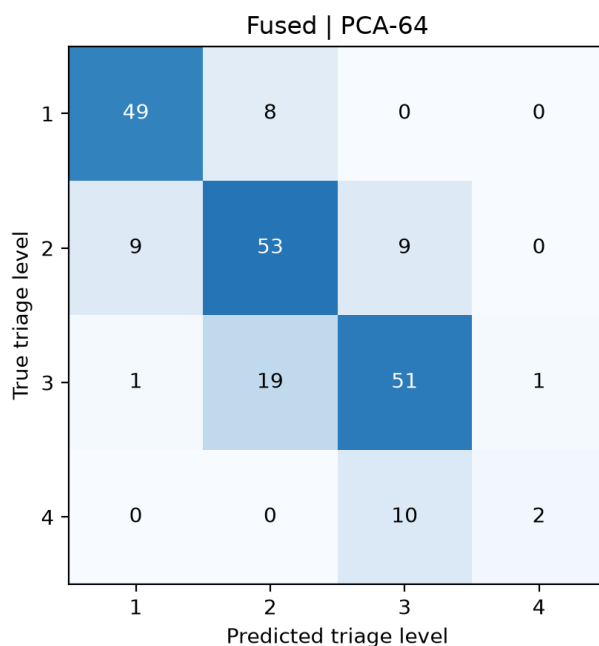
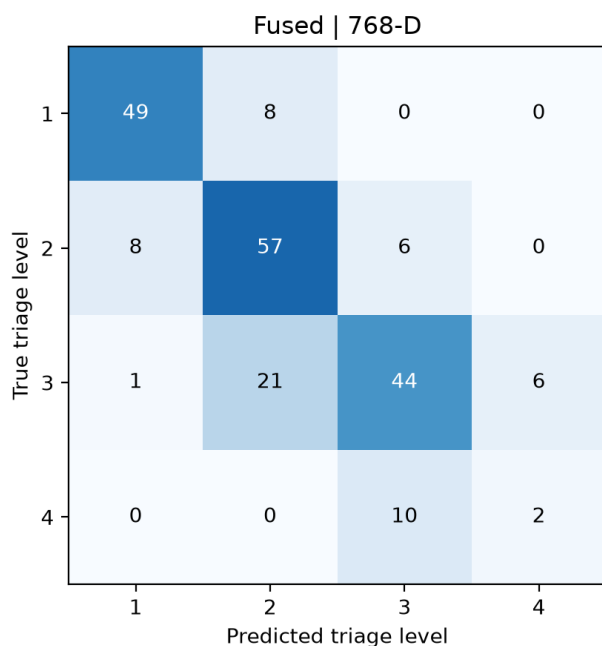
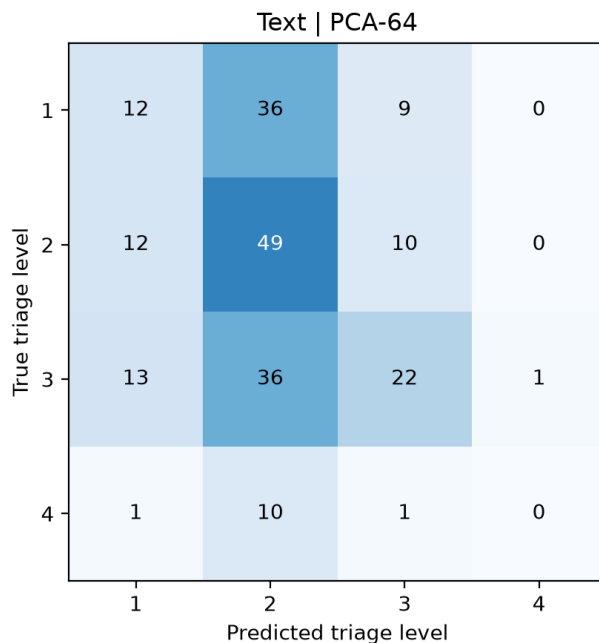
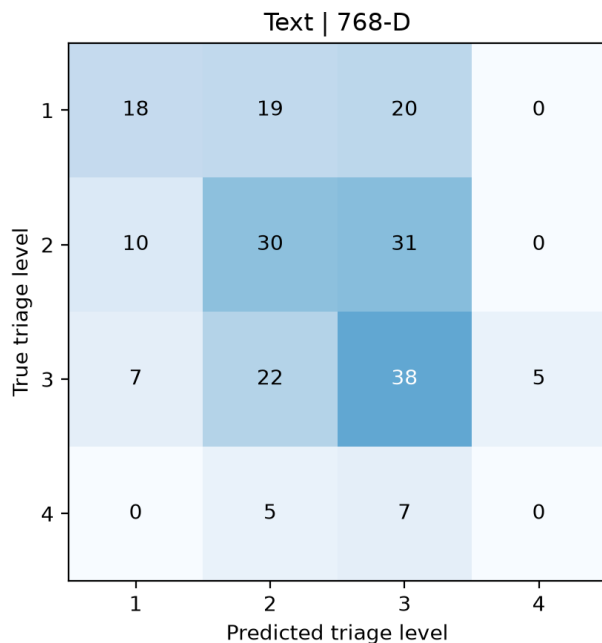
Logistic Regression: confusion matrices

Rows: true triage levels. Columns: predictions. Each matrix contains the same 212 test records. Top: text only. Bottom: text plus patient features. Left: 768-D; right: PCA-64.



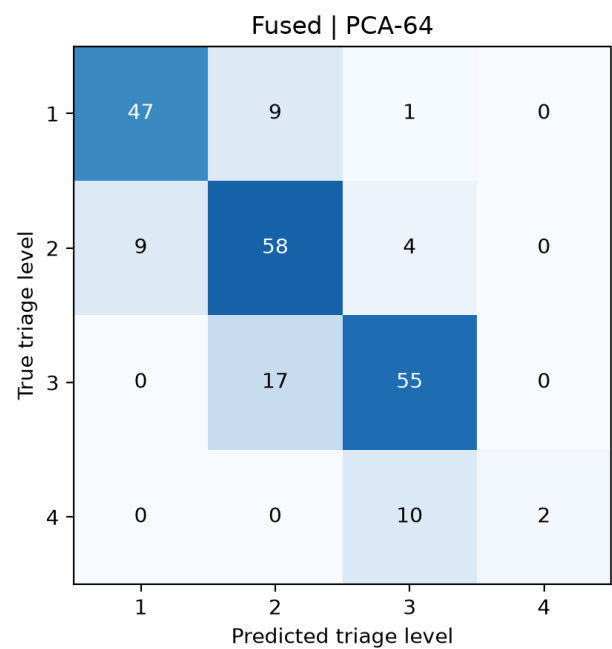
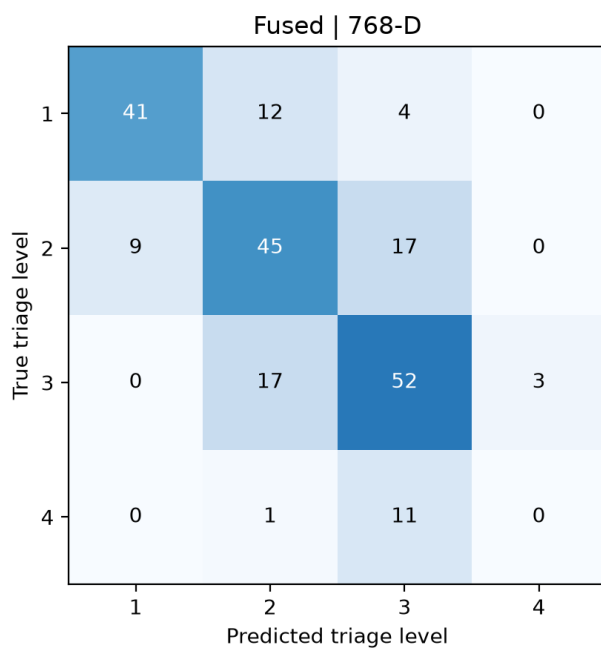
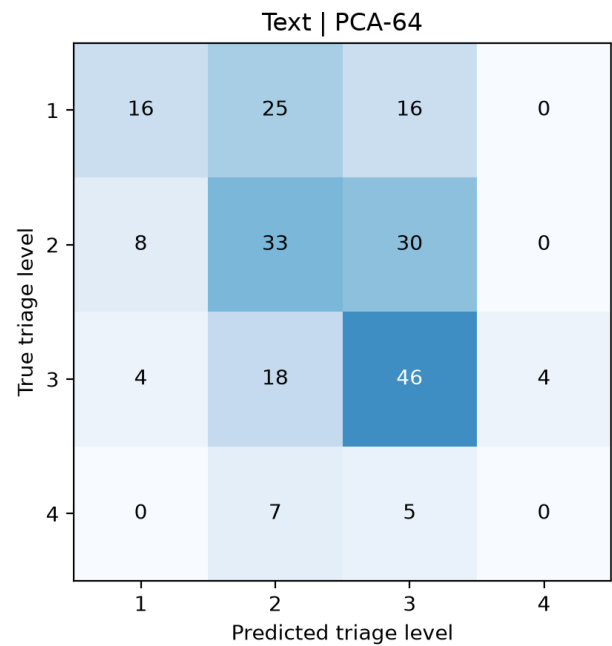
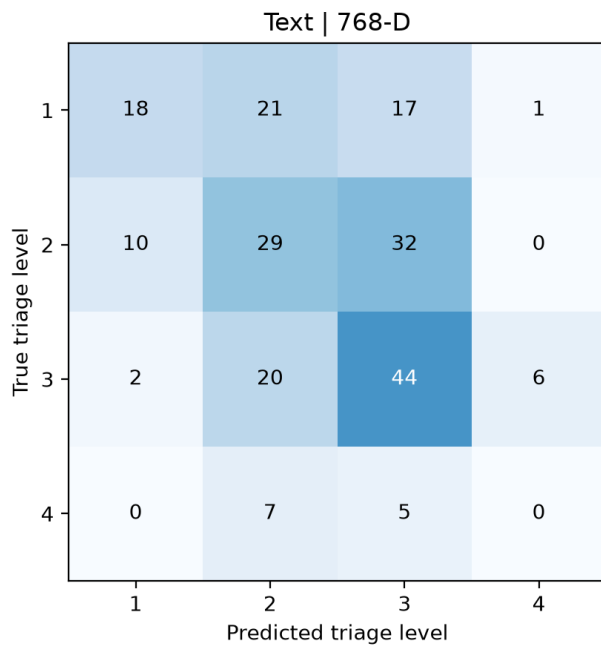
Hist Gradient Boosting: confusion matrices

Rows: true triage levels. Columns: predictions. Each matrix contains the same 212 test records. Top: text only. Bottom: text plus patient features. Left: 768-D; right: PCA-64.



Random Forest: confusion matrices

Rows: true triage levels. Columns: predictions. Each matrix contains the same 212 test records. Top: text only. Bottom: text plus patient features. Left: 768-D; right: PCA-64.



Patient-feature baseline and ordinal metrics

Patient features alone provide a reference for judging the contribution of concept text. These three models use the same 24 structured features and grouped holdout as the text comparisons.

Classifier	Accuracy	Precision*	Recall*	F1*
Logistic Regression	73.11%	67.92%	70.44%	68.60%
Hist Gradient Boosting	72.64%	63.85%	59.59%	59.71%
Random Forest	73.11%	74.04%	63.39%	65.54%

Ordinal metrics for all comparisons

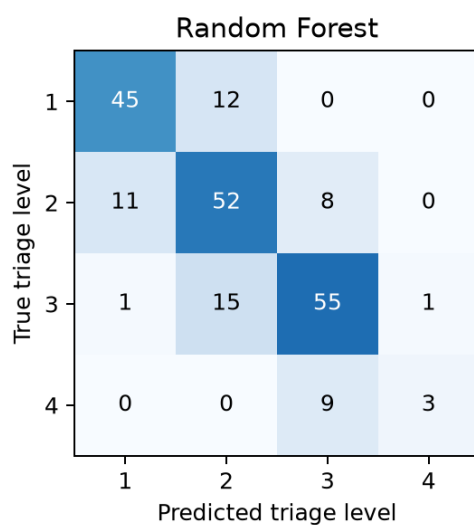
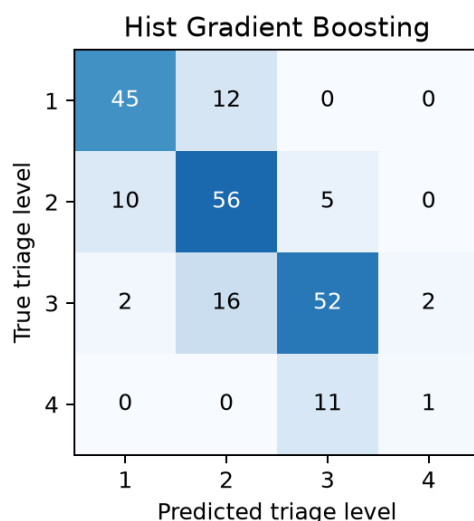
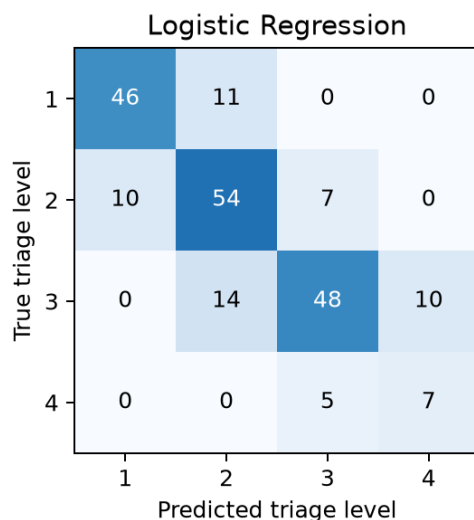
Lower triage numbers indicate greater urgency. Under-triage assigns a higher number than the true label; over-triage assigns a lower number. QWK is quadratic weighted kappa. MAE is the mean absolute error in triage levels.

Input / classifier	QWK	MAE	Under-triage	Over-triage
text_768+logreg	0.365	0.844	42.92%	19.81%
text_768+hgb	0.265	0.745	35.38%	24.06%
text_768+rf	0.316	0.703	36.32%	20.75%
fused_768+logreg	0.837	0.269	13.21%	13.68%
fused_768+hgb	0.803	0.288	9.43%	18.87%
fused_768+rf	0.711	0.373	16.98%	17.92%
text_64+logreg	0.379	0.868	45.28%	20.28%
text_64+hgb	0.100	0.769	26.42%	34.43%
text_64+rf	0.316	0.679	35.38%	19.81%
fused_64+logreg	0.826	0.278	13.21%	14.15%
fused_64+hgb	0.808	0.274	8.49%	18.40%
fused_64+rf	0.826	0.241	6.60%	16.98%
structured+logreg	0.836	0.269	13.21%	13.68%
structured+hgb	0.792	0.283	8.96%	18.40%
structured+rf	0.809	0.274	9.91%	16.98%

Key: text = concept embeddings; fused = embeddings plus patient features; structured = patient features only. logreg = Logistic Regression; hgb = Hist Gradient Boosting; rf = Random Forest.

Patient-feature baseline: confusion matrices

The same 212 test records are evaluated using patient features alone. Rows show true labels and columns show predictions.



Cross-validation and reproducibility

The uploaded script reports five-fold stratified cross-validation on the training matrices. The same procedure is used for added comparisons. PCA and scaling were fitted on the entire holdout-training partition before CV, and CV folds are not grouped by concept. These CV scores are exploratory; they are not leakage-free grouped validation.

Input / classifier	Folds	Mean macro F1	Standard deviation
text_768+logreg	5	36.35%	4.21%
text_768+hgb	5	30.53%	3.77%
text_768+rf	5	35.89%	6.50%
fused_768+logreg	5	74.56%	2.79%
fused_768+hgb	5	64.37%	4.38%
fused_768+rf	5	52.76%	7.93%
text_64+logreg	5	34.85%	2.98%
text_64+hgb	5	28.51%	3.71%
text_64+rf	5	36.35%	6.70%
fused_64+logreg	5	73.25%	2.39%
fused_64+hgb	5	66.33%	3.25%
fused_64+rf	5	66.46%	6.59%
structured+logreg	5	72.84%	3.24%
structured+hgb	5	71.15%	5.37%
structured+rf	5	74.93%	4.30%

Settings and scope

Logistic Regression: max_iter=2000, class_weight=balanced, random_state=42. Hist Gradient Boosting: max_iter=300, learning_rate=0.1, default unweighted loss, random_state=42. Added Random Forest: 300 trees, balanced class weights, random_state=42, n_jobs=2; remaining estimator settings use library defaults.

The uploaded script's six original PCA-64 / patient-feature results are reused from captured predictions. Additional 768-D and Random Forest comparisons are fitted separately. No deployed classifier is replaced. Exact source SHA-256: 5d3b9b5948d988533ab37d4d91d40b3303403aa9f47a467d2e0a021e9a9008ab.

Interpretation and Section 3.6

This is a single-split synthetic-data comparison, not clinical validation or an evaluation of live translation. Matching by exact concept string does not guarantee separation of all semantically similar complaints. Comparing many models on the holdout does not provide an untouched final validation set. Section 3.6 literature comparison remains pending because no verified comparator list and matched evaluation protocol were provided; no scores are invented.

Effect of dimensionality reduction and fusion

Differences below are percentage points. PCA change = PCA-64 minus 768-D. Fusion contribution = patient features plus concept text minus patient features alone. Positive values indicate a higher measured score on this holdout.

PCA-64 versus original 768-D

Inputs / classifier	Accuracy change	Macro F1 change
text / logreg	-2.83	-2.48
text / hgb	-1.42	-3.20
text / rf	+1.89	+0.65
fused / logreg	-0.47	-0.43
fused / hgb	+1.42	+2.17
fused / rf	+11.32	+14.78

Contribution of concept text to patient features

Text size / classifier	Accuracy change	Macro F1 change
768 / logreg	+0.00	-0.24
768 / hgb	-0.94	+1.42
768 / rf	-8.02	-14.31
64 / logreg	-0.47	-0.67
64 / hgb	+0.47	+3.59
64 / rf	+3.30	+0.47

These are observed differences from one split, without confidence intervals or significance testing. They should not be interpreted as established differences in clinical performance.

Text only: 768-D | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	54.55%	42.11%	47.52%	57
2	45.90%	39.44%	42.42%	71
3	40.00%	27.78%	32.79%	72
4	12.28%	58.33%	20.29%	12
macro avg	38.18%	41.91%	35.76%	212
weighted avg	44.32%	37.26%	39.27%	212

Accuracy: 37.26%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	51.43%	31.58%	39.13%	57
2	39.47%	42.25%	40.82%	71
3	39.58%	52.78%	45.24%	72
4	0.00%	0.00%	0.00%	12
macro avg	32.62%	31.65%	31.30%	212
weighted avg	40.49%	40.57%	39.55%	212

Accuracy: 40.57%

Random Forest

Class / average	Precision	Recall	F1	Support
1	60.00%	31.58%	41.38%	57
2	37.66%	40.85%	39.19%	71
3	44.90%	61.11%	51.76%	72
4	0.00%	0.00%	0.00%	12
macro avg	35.64%	33.38%	33.08%	212
weighted avg	43.99%	42.92%	41.83%	212

Accuracy: 42.92%

Text only: PCA-64 | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	53.33%	42.11%	47.06%	57
2	44.07%	36.62%	40.00%	71
3	34.78%	22.22%	27.12%	72
4	11.29%	58.33%	18.92%	12
macro avg	35.87%	39.82%	33.27%	212
weighted avg	41.55%	34.43%	36.33%	212

Accuracy: 34.43%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	31.58%	21.05%	25.26%	57
2	37.40%	69.01%	48.51%	71
3	52.38%	30.56%	38.60%	72
4	0.00%	0.00%	0.00%	12
macro avg	30.34%	30.16%	28.09%	212
weighted avg	38.81%	39.15%	36.15%	212

Accuracy: 39.15%

Random Forest

Class / average	Precision	Recall	F1	Support
1	57.14%	28.07%	37.65%	57
2	39.76%	46.48%	42.86%	71
3	47.42%	63.89%	54.44%	72
4	0.00%	0.00%	0.00%	12
macro avg	36.08%	34.61%	33.74%	212
weighted avg	44.79%	44.81%	42.96%	212

Accuracy: 44.81%

Patient features + text: 768-D | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	83.93%	82.46%	83.19%	57
2	68.75%	77.46%	72.85%	71
3	79.31%	63.89%	70.77%	72
4	38.89%	58.33%	46.67%	12
macro avg	67.72%	70.54%	68.37%	212
weighted avg	74.73%	73.11%	73.44%	212

Accuracy: 73.11%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	84.48%	85.96%	85.22%	57
2	66.28%	80.28%	72.61%	71
3	73.33%	61.11%	66.67%	72
4	25.00%	16.67%	20.00%	12
macro avg	62.27%	61.01%	61.12%	212
weighted avg	71.23%	71.70%	71.00%	212

Accuracy: 71.70%

Random Forest

Class / average	Precision	Recall	F1	Support
1	82.00%	71.93%	76.64%	57
2	60.00%	63.38%	61.64%	71
3	61.90%	72.22%	66.67%	72
4	0.00%	0.00%	0.00%	12
macro avg	50.98%	51.88%	51.24%	212
weighted avg	63.17%	65.09%	63.89%	212

Accuracy: 65.09%

Patient features + text: PCA-64 | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	82.46%	82.46%	82.46%	57
2	68.75%	77.46%	72.85%	71
3	78.95%	62.50%	69.77%	72
4	38.89%	58.33%	46.67%	12
macro avg	67.26%	70.19%	67.93%	212
weighted avg	74.21%	72.64%	72.90%	212

Accuracy: 72.64%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	83.05%	85.96%	84.48%	57
2	66.25%	74.65%	70.20%	71
3	72.86%	70.83%	71.83%	72
4	66.67%	16.67%	26.67%	12
macro avg	72.21%	62.03%	63.29%	212
weighted avg	73.03%	73.11%	72.13%	212

Accuracy: 73.11%

Random Forest

Class / average	Precision	Recall	F1	Support
1	83.93%	82.46%	83.19%	57
2	69.05%	81.69%	74.84%	71
3	78.57%	76.39%	77.46%	72
4	100.00%	16.67%	28.57%	12
macro avg	82.89%	64.30%	66.02%	212
weighted avg	78.04%	76.42%	75.36%	212

Accuracy: 76.42%

Patient features only | class details

All values below are measured on the same 212 held-out records. Precision, recall and F1 are shown as percentages. Support is the number of true records in the class.

Logistic Regression

Class / average	Precision	Recall	F1	Support
1	82.14%	80.70%	81.42%	57
2	68.35%	76.06%	72.00%	71
3	80.00%	66.67%	72.73%	72
4	41.18%	58.33%	48.28%	12
macro avg	67.92%	70.44%	68.60%	212
weighted avg	74.48%	73.11%	73.44%	212

Accuracy: 73.11%

Hist Gradient Boosting

Class / average	Precision	Recall	F1	Support
1	78.95%	78.95%	78.95%	57
2	66.67%	78.87%	72.26%	71
3	76.47%	72.22%	74.29%	72
4	33.33%	8.33%	13.33%	12
macro avg	63.85%	59.59%	59.71%	212
weighted avg	71.41%	72.64%	71.41%	212

Accuracy: 72.64%

Random Forest

Class / average	Precision	Recall	F1	Support
1	78.95%	78.95%	78.95%	57
2	65.82%	73.24%	69.33%	71
3	76.39%	76.39%	76.39%	72
4	75.00%	25.00%	37.50%	12
macro avg	74.04%	63.39%	65.54%	212
weighted avg	73.46%	73.11%	72.51%	212

Accuracy: 73.11%